# HARIHARAN

**NPD Engineer · Industrial Automation · PLC & SCADA**
**Robotics Engineer | ECN & ECO | SAP | BOM**

Puducherry, India · +91 8056409115 · hariharanrae@gmail.com · LinkedIn

## Professional Summary

Engineering professional with hands-on experience in Project Coordination Micro Data Center systems, industrial automation, and electrical infrastructure. Skilled in PLC programming, HMI development, AutoCAD Electrical, and SAP-based BOM management. Strong expertise in troubleshooting, installation, and commissioning, along with experience in BOM costing and techno-commercial support, working closely with production to technical alignment.

## Core Competencies

**Automation & Maintenance**
- PLC Programming (Ladder Logic/FBD)
- Robotics & Automation
- Industrial Machine Maintenance
- Electrical Maintenance
- HMI & SCADA Design

**Electrical & Control Systems**
- PDU & UPS
- VFD & Control Circuits
- MATLAB
- Power Distribution Systems
- OCP ORv3 Infrastructure
- Modbus RTU | Modbus TCP

**Operations & Compliance**
- DCIM & SNMP Monitoring
- ECN & ECO Workflows
- SAP (BOM, PR, PO)
- FAT & UL Compliance
- Commercial

## Software Proficiencies

| | |
|---|---|
| **Electrical Design:** | AutoCAD Electrical |
| **PLC & HMI Tools:** | PLC Next Engineer · PC Worx · GX Works 2 & 3 · JMobile Studio |
| **Robotics:** | RoboDK · Robo Studio · Dobot Studio |

## Professional Experience

**Eaton Power Quality Pvt. Ltd.** *Jun 2025 – Present*
*Graduate Engineer Trainee · New Product Development (Data Center Solutions)*
- **Automation:** Developed Micro Data Center infrastructure; programmed PLC logic and HMI screens for system monitoring using RTU and TCP protocols.
- **Maintenance & Testing:** Conducted electrical maintenance and Factory Acceptance Testing (FAT) for OCP ORv3 systems.
- **Compliance:** Collaborated with sales teams to develop customer-specific costing, bidding, and techno-commercial solutions aligned with project requirements.
- **Operations:** Managed SAP BOMs for NPD programs ensuring technical accuracy; assisted services teams during site commissioning.

**Cornerstone engineering Solutions pvt ltd.**

*Application Engineer Intern*
- **Automation:** Supported industrial automation projects by assisting in implementation of control systems and customized solutions based on client requirements.
- **Installation & Commissioning:** Assisted in installation, commissioning, and on-site setup of automation systems across industrial environments.
- **Service & Troubleshooting:** Provided support in service activities, basic troubleshooting, and maintenance of automation equipment to ensure operational efficiency.
- **Operations**: Coordinated with internal teams to support consultancy-based project execution and ensure timely delivery of engineering solutions.

## Education

**B.Tech · Robotics and Automation** *2021 – 2025*
Manakula Vinayagar Institute of Technology, Puducherry
**Higher Secondary Education (HSE)** *2019 – 2021*
SKY matric Hr Sec School, Tiruvannamalai
**SSLC** *2018 – 2019*
SKY matric Hr Sec School, Tiruvannamalai

## Projects & Publications

**Gesture-Controlled Autonomous Robot with Manipulator | Academic Project**
- Developed an IoT-enabled autonomous mobile robot with a robotic manipulator using Arduino and ESP32.
- Implemented gesture-based control for robot navigation and pick-and-place operations.
- Integrated controller, sensors and actuators for coordinated robotic movement and manipulation.

**Design of Push-Pull Meter System | Consultancy Project**
- Designed and developed a push-pull meter for industrial quality testing, supporting implementation from design to installation.

## Awards & Certifications

### Certification
- **Siemens Mobility**: Operations Industrial Engineer Job Simulation.
- **UiPath Academy:** RPA Citizen Developer Foundation Training.
- **Kuka Robots:** Hands-on industrial Automation Workshop.
- **ADSC INDIA:** Advance Electricals.
- **Infosys:** Robotics course.

### Awards
- **Best Alumni Award:** continuous contributions to national level hackathons & workshops.
- **Codefest Hackathon:** second price at 24 hours national level Hackathon.
- **DSTE Puducherry:** Second price at National level E-Waste Hackathon.
- **SAE INDIA:** First Price at National level Mobile Robotics competition.
- **SCIMIT:** First price at National level science day celebration.

## Strategic Areas of Interest

- Data **Center Infrastructure:** OCP ORv3 Rack Standards, **PDU (Power Distribution Units), UPS Systems, Cooling Systems**, Liquid Cooling Technologies, Modular Power Units, and Data Center Automation.
- Robotics **& Industrial Automation:** IIoT Integration, Industrial Robotics, PLC-based Machine Maintenance, and Automated Quality Testing.
- Power **Electronics:** Smart PDU Design, Energy Efficient Power Distribution, and Renewable Energy Integration.
- Techno-**Commercial & Sales Engineering:** Costing, Bidding, Proposal Support, BOM Price Estimation, and Customer Requirement Analysis.
- Safety **& Compliance:** UL/IEC Standards, Product Validation Testing, and Engineering Compliance.
- Diagnostics **& Troubleshooting:** Industrial Sensors, Controllers, System Root Cause Analysis, and Reliability Improvement.